Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 14: Logistic Regression

Sources

- **Machine Learning for Intelligent Systems**, Kilian Weinberger, Cornell, Video Lectures 35-37, https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lect_urenote20.html
- **Deep Learning Specialization, Andrew Ng** https://www.coursera.org/specializations/deep-learning?utm_source=deeplearningai&utm_medium=institutions&utm_campaign=SocialYoutubeDLSC1W1L1
  - **Video Lectures: C1W3L1 – C1W4L6:** https://www.youtube.com/playlist?list=PLpFsSf5Dm-pd5d3rjNtIXUHT- v7bdaEIe
- **A beginner's guide to deriving and implementing backpropagation,** by Pranav Budhwant https://link.medium.com/Zp3zxNWpf6
- **Tensorflow playground:** https://playground.tensorflow.org/
- **Machine Learning Playground:** https://ml-playground.com/#

An Event Occurs

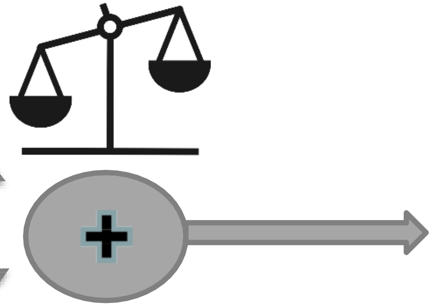
But why???

Did not study w_1

Distracting kids w_2

Noisy house w_3

Its the fault of his teacher, his life, the universe and everything w_4



George has failed ML this semester!

This is what logistic regression tries to capture and quantify!

Logistic Regression

- Discover the relation between features/cues/events and some outcome
- Logistic regression is the baseline supervised machine learning algorithm for classification
- It has a very close relationship with neural networks
  - A neural network can be viewed as a series of logistic regression classifiers stacked on top of each other
- The outcome is viewed as a weighted sum of features
- A bias (θ_0, or y-intercept) to adjust the trend of the sum
- The learned weights enhance important (discriminating) features and suppress unimportant ones

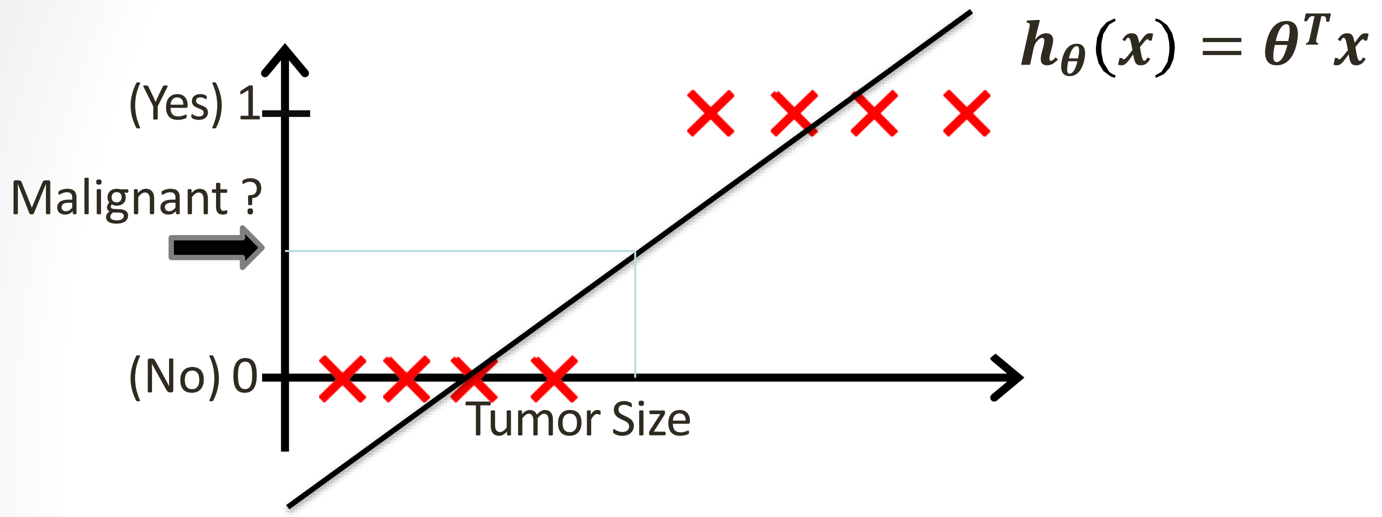
Classification

Classify an observation into:

- **One of two classes (binary or binomial)**
  - Sentiment: positive / negative?
  - Email: Spam / Not Spam?
  - Online Transactions: Fraudulent (Yes / No)?
  - Tumor: Malignant / Benign ?

- **One of many classes (multinomial)**
  - Sentiment: Positive / negative / neutral?
  - Emotion: Happy, Sad, Surprised, Angry,…?
  - Part-of-Speech tag: Noun / verb / adjective / adverb /…?
  - Recognize a word: One of |V| tags

Can we use regression for classification?

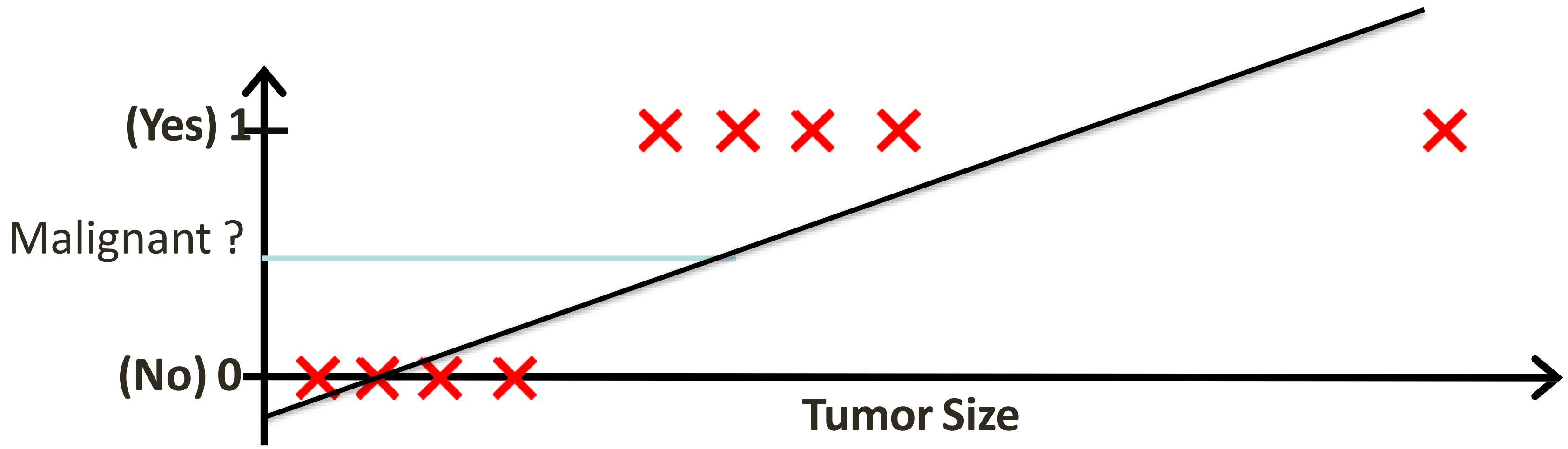


Threshold classifier output $h_\theta(x)$ at 0.5:

If $h_\theta(x) \geq 0.5$, predict "y = 1"

If $h_\theta(x) < 0.5$, predict "y = 0"

Can we use regression for classification?



Threshold classifier output $h_\theta(x)$ at 0.5:

If $h_\theta(x) \geq 0.5$, predict "y = 1"

If $h_\theta(x) < 0.5$, predict "y = 0"

Logistic Regression

- Logistic regression learns a vector of **weights,** $\boldsymbol{\theta_i}$, from a training set
- Each weight $\boldsymbol{\theta_i} \in \mathbb{R}$, is associated with one of the input features $\boldsymbol{x_i}$
- The weight $\boldsymbol{\theta_i}$ represents how important $\boldsymbol{x_i}$ is to the classification decision
  - can be positive (meaning the feature is associated with the class)
  - or negative (meaning the feature is not associated with the class)

- E.g. “positive sentiment” versus “negative sentiment”
  - *The features represent counts of words in a document*
  - $P(y = 1|x)$ is the probability that the document has positive sentiment
  - $P(y = 0|x)$ is the probability that the document has negative sentiment.
  - *“awesome”* has a high positive weight
  - *“abysmal”* has a very negative weight

$$\boldsymbol{h_\theta}(\ \boldsymbol{x}) = \boldsymbol{\theta^T x}$$

“Squishing” results to be between 0 and 1

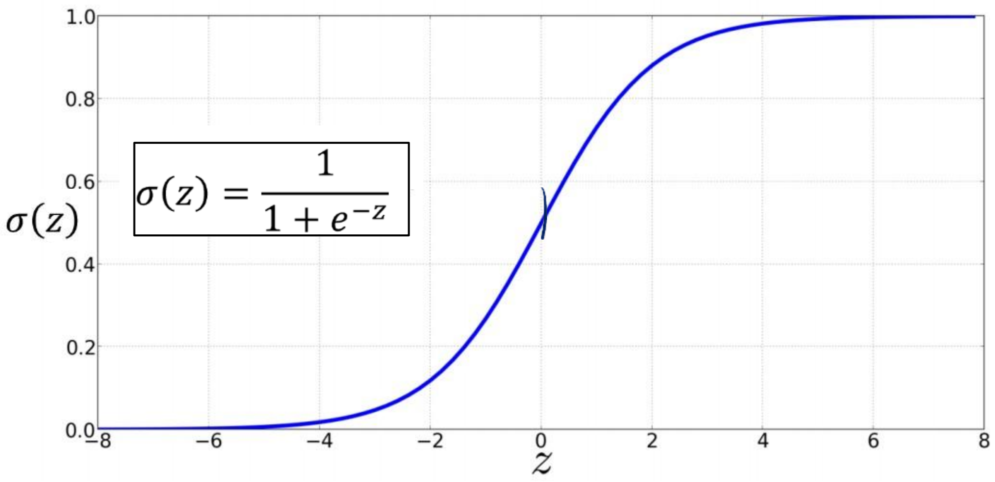
- $\boldsymbol{h\theta(x)}$ is not forced to be a legal probability, that is, to lie between 0 and 1
- In fact, since weights are real-valued, the output might even be negative
- $\boldsymbol{h_\theta}(\ \boldsymbol{x})$ ranges from $-\infty$ to ∞.
- To create a probability, we’ll pass $\boldsymbol{h_\theta}(\ \boldsymbol{x})$ through the **sigmoid** function, $\sigma(z)$.
  - The sigmoid function (named because it looks like an *s*) is also called the **logistic function.**

$$\sigma(Z) = \frac{1}{1 + \ e^{-z}}$$

$$\sigma(Z) = \frac{1}{1 + \ e^{-h_\vartheta(x)}}$$

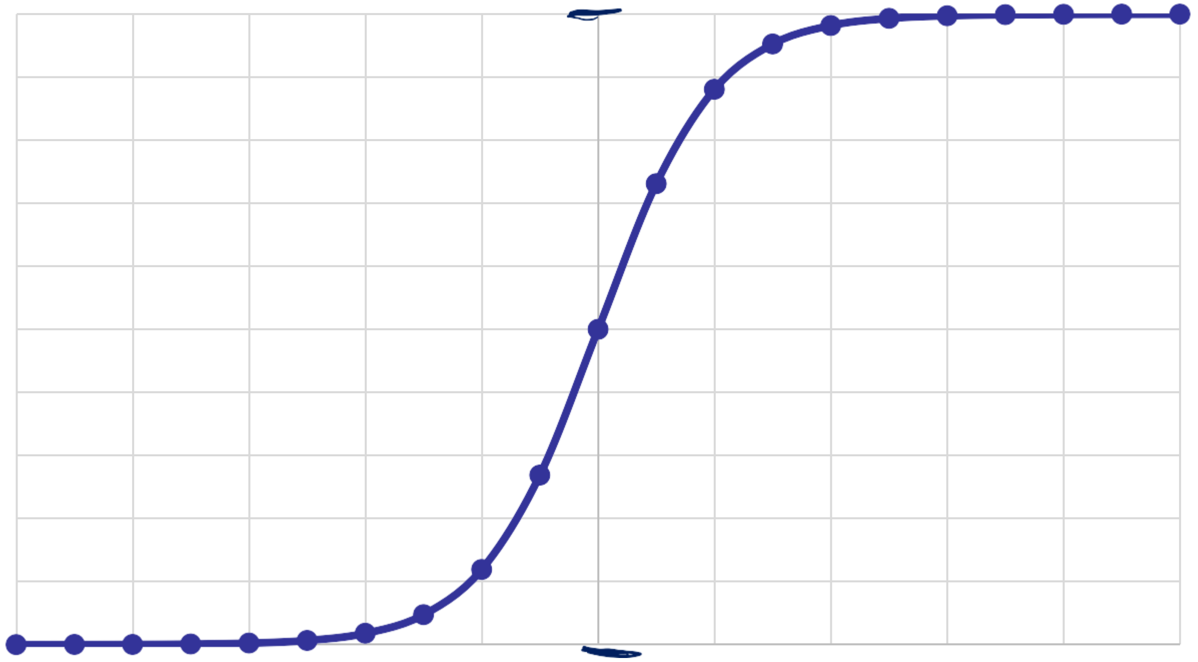
$$z = \theta^T x + b$$

$$h(x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$



The Sigmoid

x	Sigmoid
-10	4.53979E-05
-9	0.000123395
-8	0.00033535
-7	0.000911051
-6	0.002472623
-5	0.006692851
-4	0.01798621
-3	0.047425873
-2	0.119202922
-1	0.268941421
0	0.5
1	0.731058579
2	0.880797078
3	0.952574127
4	0.98201379
5	0.993307149
6	0.997527377
7	0.999088949
8	0.99966465
9	0.999876605
10	0.999954602



Advantages of a Sigmoid

- Maps real-valued numbers (ℝ) into the range [0,1]
- Nearly linear around 0 but has a sharp slope toward the ends, it tends to squash outlier values toward 0 or 1
- It is differentiable, which is be handy for learning
- To make it a probability, we just need to make sure that the two cases, *p*(*y* = 1) and *p*(*y* = 0), sum to 1:

$$P(y == 1) = \sigma(\vartheta^T x)$$
$$= \frac{1}{1 + e^{-\vartheta^T x}}$$

$$P(y == 0) = 1 - \sigma(\vartheta^T x)$$
$$= \frac{e^{-\vartheta^T x}}{1 + e^{-\vartheta^T x}}$$

- How do we decide?
  - For a test instance *x*, we say **yes** if the probability *P*(*y* = 1|*x*) is more than .5, and **no** otherwise.
  - We call .5 the **decision boundary**

Example Sentiment Classification

- Binary sentiment classification on movie review text
- 6 features $x_1 ... x_6$ of the input
- Learned weights for each of these features: [2.5,−5.0,−1.2,0.5,2.0,0.7], while $\theta_0 = 0.1$
- Note that $w1 = 2.5$ is positive, while $w2 = -5.0$
  - negative words are negatively associated with a positive sentiment decision and are about twice as important as positive words.
  - For a test instance x, we say **yes** if the probability $P(y = 1|x)$ is more than .5, and **no** otherwise.
  - We call .5 the **decision boundary**

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

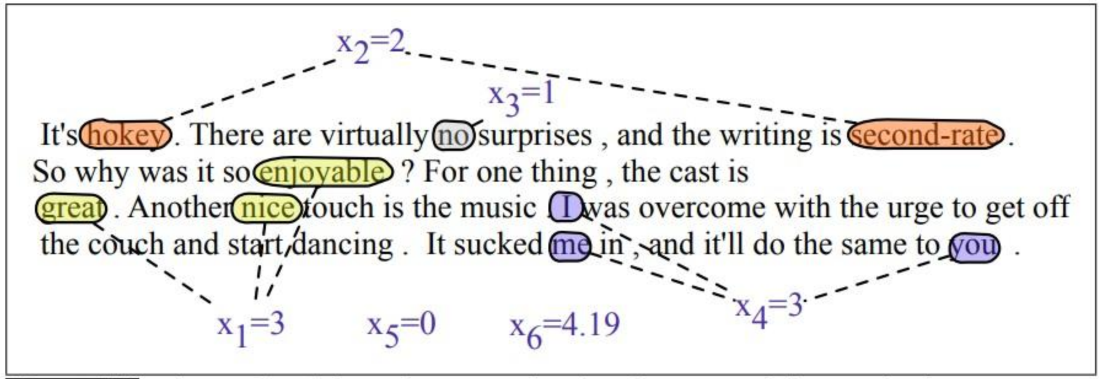


Figure 5.2 A sample mini test document showing the extracted features in the vector x.

Example Sentiment Classification

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if ``no''} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if ``!''} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(66) = 4.19$

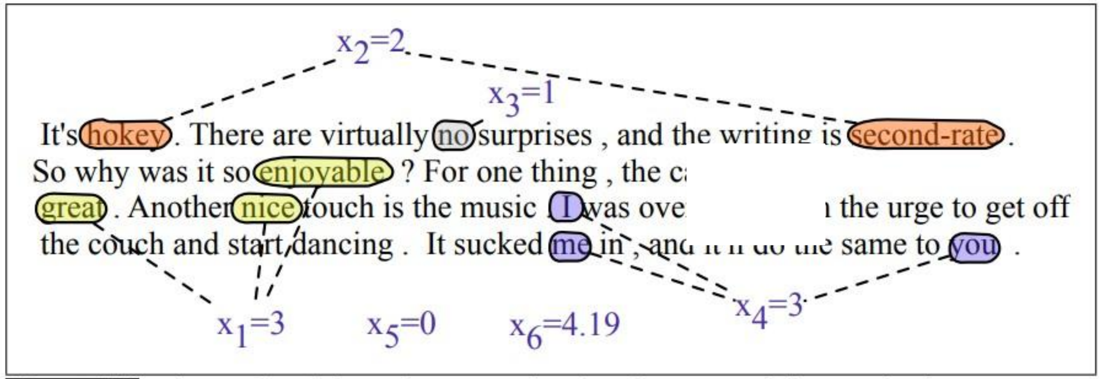


Figure 5.2 A sample mini test document showing the extracted features in the vector x.

$$
\begin{aligned}
p(+|x) = P(Y=1|x) &= \sigma(w \cdot x + b) \\
&= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\
&= \sigma(.833) \\
&= 0.70 \qquad (5.6) \\
p(-|x) = P(Y=0|x) &= 1 - \sigma(w \cdot x + b) \\
&= 0.30
\end{aligned}
$$

Thank You